

THOMAS LAUTENSCHLÄGER

Senior Software Developer & Machine Learning Engineer | M.Sc. Computer Science

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EXPERIENCE

Freelance Software Developer / ML Engineer

Self-Employed

- Aug 2022 – Present
- Darmstadt, Germany

End-to-end development and operation of production AI/ML systems for enterprise clients, from architecture to deployment, including technical leadership. Selected engagements:

AI Platform Engineer

Pharma & Medical Technology

- Built production-grade RAG and agentic-AI systems for processing regulatory content, e.g. tracking changes in regulations and legal texts, and for automating business processes such as proposal generation.
- Deployed and served multiple LLMs with GPU acceleration on Kubernetes (NVIDIA GPU Operator, vLLM, RayServe); operated a vector database (Milvus) for semantic search and retrieval; integrated RAG into chatbot systems.

ML / MLOps Engineer – Renewable Energy Forecasting

50Hertz (TSO)

- Designed and productionized an ML platform for solar and wind generation forecasting, actively operated in a high-voltage grid control center.
- Developed a real-time inference service, automated data-quality checks, and full model lifecycle management (MLflow, Azure ML); monitoring and alerting with Grafana and OpenTelemetry.
- Tech lead of a cross-functional Scrum team; owned architecture decisions and stakeholder alignment.

Machine Learning Engineer – Grid Monitoring

E.ON

- Designed and developed an automated, multi-tenant monitoring system for low-voltage grids, combining ML and rule-based fault detection; scaled for a group-wide rollout across multiple distribution system operators (Python, Kafka, Kubernetes, Azure).

Internal Python SDK for Data Scientists

Energy Sector

- Built a Python SDK for simplified sensor-data access with Azure-compliant authentication, automated CI/CD, and full documentation; established Python best practices and a mono-repo structure adopted as a reference for further projects.

Machine Learning Scientist / Quant Analyst

Inkubator 100 Plus GmbH & Co. KG

- Feb 2021 – Aug 2022
- Darmstadt, Germany
- Designed quantitative investment strategies and applied machine learning to stock market analysis; conducted backtesting and trade performance evaluation.
- Performed time series analysis (statsmodels, PyTorch) and applied probabilistic non-linear gradient solvers for risk minimization.
- Implemented a hypothesis testing framework (scikit-learn, SciPy) and a Django-based dashboard for visualizing hypotheses and backtests.

Machine Learning Engineer

Inkubator 100 Plus GmbH & Co. KG

- Feb 2020 – Feb 2021
- Darmstadt, Germany
- Developed financial data processing pipelines and data/visualization architectures; implemented backtesting software (NumPy, Pandas).
- Built a relational and multi-dimensional data lake (PostgreSQL, MongoDB) exposed via FastAPI; Django application for executing and evaluating stock trades.
- Established CI/CD with GitHub Actions and a Docker-based containerized deployment environment.

Student Research Assistant

TU Darmstadt – Laboratory of Lighting Technology

- Aug 2018 – Oct 2019
- Darmstadt, Germany
- Developed a light setting controller via non-linear optimization (TensorFlow, Python).
- Implemented state-of-the-art pupil detection algorithms for live tracking systems (C++) and integrated deep learning algorithms into a MATLAB-based UI tool.

System Administrator & DevOps

Appschmiede

- Jul 2016 – Aug 2018
- Darmstadt, Germany
- Established the company's IT infrastructure (automated backups, failover networks, firewall); built the backend and MQTT data streaming pipeline for a bicycle riding event.

SKILLS

ML & AI

Python PyTorch TensorFlow scikit-learn
Hugging Face LangChain LlamaIndex vLLM
MLflow

Infrastructure & MLOps

Docker Kubernetes OpenShift Helm ArgoCD
Terraform NVIDIA CUDA

Cloud

Azure ML/AI AWS Google Cloud

Data & Backend

FastAPI Kafka PostgreSQL TimescaleDB
Milvus

Observability

Grafana OpenTelemetry Prometheus

EDUCATION

M.Sc. Computer Science

TU Darmstadt

- April 2018 – September 2020
- **Master thesis** – Variational Inference for Switching Dynamics
 - Bayesian rARHMMs for identification and control of switching dynamics; integration into reinforcement learning
- **Focus:** reinforcement learning, optimization of static and dynamic systems, statistical machine learning

B.Sc. Computer Science

TU Darmstadt

- April 2013 – Feb 2017
- **Bachelor thesis** – Darknet Crawling and Data Analysis (extraction and analysis of market data, Scrapy)

LANGUAGES

German – native
English – business fluent
Spanish – basic

PROJECTS

Autonomous trading bot

- Live trading bot operating autonomously in the market; risk management via non-linear gradient solvers in PyTorch.

Burrolib: a Markov game framework

- Framework for simulating multi-agent Markov games with an agent interface for integrating RL algorithms (PyTorch).

Reinforcement learning on real hardware

- Implemented and evaluated recent RL algorithms on Quanser environments (Furuta pendulum, cart-pole) using PyTorch.

ACHIEVEMENTS

- Awarded as one of the top students of the reinforcement learning triple (lecture, project, seminar) at the Intelligent Autonomous Systems Group.